BMSCE DATA STRUCTURES

Unit 1 & Unit 2 — Exam-Oriented Question Bank

Scope: Unit 1 and Unit 2 only up to queues. Topics after queues (including linked lists) are excluded.

Question formats: 5-mark, 10-mark and 20-mark questions, including subparts, algorithms, tracing and C-programming questions.

Prepared as a revision and practice resource. The online-paper patterns discussed in the accompanying chat are indicative, not a guarantee of the current exam.

How to Use This Question Bank

1. Start with the 5-mark questions for definitions, comparisons and short explanations.
2. Practise every 10-mark question with clear subpart-wise answers and algorithms.
3. Attempt the 20-mark questions under timed conditions; show working for expression conversion, recursion and queue traces.
4. For queue questions, always state the indexing convention and the conditions for full and empty queues.

Exam-pattern note: The question-paper layout discussed earlier appeared to include a 10-mark Part A and two 20-mark sections with choices/subparts. Confirm the exact marks distribution against your current faculty-provided format.

Part I — 5-Mark Questions

Unit 1: Introduction to Data Structures, Stacks and Recursion

Q1. Differentiate between **data** and **information**, with examples.

Q2. Classify data structures. Explain the major categories with suitable examples.

Q3. Differentiate between **primitive** and **non-primitive** data structures.

Q4. Differentiate between **data structures** and **file organization**.

Q5. Explain the fundamental operations on data structures: traversal, insertion, deletion, searching, sorting and merging.

Q6. Define a stack. Explain the **LIFO** principle with an example.

Q7. Explain the stack operations: push, pop, peek/top, isEmpty and isFull.

Q8. Explain array representation of a stack. State the initial value of TOP and the overflow and underflow conditions.

Q9. What is recursion? Explain the base case and recursive case.

Q10. What is a structure in C? Explain how structure members are accessed.

Unit 2: Queues

Q11. Define a queue and explain the **FIFO** principle.

Q12. Explain the basic queue operations: enqueue, dequeue, front/peek, isEmpty and isFull.

Q13. Differentiate between a linear queue and a circular queue.

Q14. Explain the advantages of a circular queue over a linear queue.

Q15. Explain array representation of a queue, including overflow and underflow conditions.

Q16. What is a priority queue? Differentiate between ascending and descending priority queues.

Q17. List and explain applications of a priority queue.

Q18. Define a deque. How is it different from a standard queue?

Q19. Differentiate between input-restricted and output-restricted deques.

Q20. List and explain applications of queues.

Q21. State the full and empty conditions for a circular queue, assuming the one-empty-slot convention.

Q22. Differentiate between a stack and a queue.

Part II — 10-Mark Questions

Unit 1: Stacks, Expression Conversion, Evaluation and Recursion

Q23. Convert infix to postfix

(a) Write an algorithm to convert an infix expression into postfix form. (5 marks)

(b) Convert `A+B*(C-D)` into postfix, showing the stack and output after each scanned symbol. (5 marks)

Q24. Postfix evaluation

(a) Write an algorithm to evaluate a postfix expression using a stack. (5 marks)

(b) Evaluate `23*54*+9-`, showing the stack after each operation. (5 marks)

Q25. Stack operations

(a) Write algorithms for PUSH and POP using an array representation of a stack, including overflow and underflow checks. (5 marks)

(b) Trace the operations: PUSH(10), PUSH(20), POP, PUSH(30), POP. Show the stack after each operation. (5 marks)

Q26. Recursion and tracing

(a) Define recursion and explain the purpose of a base case and a recursive case. (4 marks)

(b) Trace the following function for `fun(4,3)` and show the sequence of calls and the returned result. (6 marks)

```
int fun(int a, int b) {
if (b == 0) return 0;
if (b % 2 == 0) return fun(a + a, b / 2);
return fun(a + a, b / 2) + a;
}
```

Q27. Data-structure fundamentals

(a) Classify data structures with examples. (5 marks)

(b) Explain any five fundamental operations performed on data structures. (5 marks)

Q28. Linear queue operations

(a) Write algorithms for ENQUEUE and DEQUEUE for a linear queue implemented using an array. (5 marks)

(b) Trace a given sequence of queue operations and show FRONT and REAR after each operation. (5 marks)

Q29. Circular queue

(a) State the full and empty conditions for an array-based circular queue. (4 marks)

(b) Write insertion and deletion algorithms for a circular queue using modulo arithmetic. (6 marks)

Q30. Priority queue

(a) Explain ascending and descending priority queues with examples. (4 marks)

(b) Write algorithms for insertion, deletion and display for a priority queue. (6 marks)

Q31. Deque

(a) Define a deque and explain input-restricted and output-restricted deques. (4 marks)

(b) Write algorithms for insertion and deletion at both ends of a deque implemented with a circular array. (6 marks)

Q32. Linear versus circular queue

(a) Compare linear and circular queues. (5 marks)

(b) Explain false overflow in a linear queue using an example where earlier positions become unused after deletions. (5 marks)

Part III — 20-Mark Questions

Unit 1: Integrated Long-Answer Questions

Q33. Stack and expression conversion

(a) Explain stack applications and array representation of a stack. (4 marks)

(b) Write an algorithm to convert infix to postfix. (6 marks)

(c) Convert `A+B*(C-D)^E` to postfix, showing the stack steps. (5 marks)

(d) State the advantages of postfix expressions. (5 marks)

Q34. Postfix evaluation and recursion

(a) Write an algorithm for postfix evaluation using a stack. (5 marks)

(b) Evaluate `12 10 - 9 6 + *`, showing each stack state. (5 marks)

(c) Explain recursion and its essential components. (4 marks)

(d) Trace a recursive function, showing calls and return values. (6 marks)

Q35. Introduction to data structures

(a) Define data, information and data structure. (4 marks)

(b) Classify data structures and give suitable examples. (6 marks)

(c) Differentiate data structures from file organization. (5 marks)

(d) Explain the fundamental operations on data structures. (5 marks)

Unit 2: Queue-Based Long-Answer Questions

Q36. Circular queue: algorithms and trace

(a) Explain the limitations of a linear queue and the advantages of a circular queue. (5 marks)

(b) Write algorithms for insertion and deletion in a circular queue. (7 marks)

(c) A circular queue has capacity 5. Initially FRONT = 1 and REAR = 3, with 10, 20 and 30 stored in FIFO order in positions 1, 2 and 3. Insert 40, insert 50, attempt to insert 60, delete twice, insert 70 and insert 80. Show the array, FRONT, REAR and queue status after every operation. State the indexing convention you use. (8 marks)

Q37. Linear queue and C implementation

(a) Define a queue and explain its operations and array representation. (5 marks)

(b) Write a menu-driven C program for a linear queue with enqueue, dequeue, display and exit options. Handle overflow and underflow. (10 marks)

(c) Explain five applications of queues. (5 marks)

Q38. Priority queue

(a) Define a priority queue and distinguish min-priority/ascending from max-priority/descending queues. (4 marks)

(b) Write algorithms for insertion, deletion and display for a priority queue. (10 marks)

(c) Explain how equal priorities may be handled and list six applications of priority queues. (6 marks)

Q39. Deque

(a) Define a deque and explain its types with examples. (6 marks)

(b) Write an algorithm for insertion at the front and rear of a deque. (6 marks)

(c) Write algorithms for deletion at the front and rear, including empty/full edge cases. (8 marks)

Q40. Queue types and comparisons

(a) Compare linear and circular queues. (6 marks)

(b) Compare simple queues, priority queues and deques. (8 marks)

(c) Choose a suitable queue type for a printer queue, an emergency-room triage system and a system requiring insertion/deletion at both ends. Justify each choice. (6 marks)

Part IV — Extra Practice Drill

A. Infix-to-Postfix Conversion

Infix expression	Postfix answer
`A+B*C-D`	`ABC*+D-`
`(A+B)*(C-D)`	`AB+CD-*`
`A*(B+C)/D`	`ABC+*D/`
`(A+B/C)`	`ABC/+`
`A+B*(C-D)/E`	`ABCD-*E/+`

B. Additional Practice Questions

Q1. Evaluate postfix expression `5 6 2 + * 12 4 / -`.

Q2. Evaluate postfix expression `8 2 / 3 4 * +`.

Q3. Write a recursive C function to calculate factorial of a non-negative integer.

Q4. Write a recursive C function to find the sum of digits of an integer.

Q5. Trace recursive calls and return values for a supplied recursive function.

Q6. Write a menu-driven C program for stack operations using an array.

Q7. Trace a stack sequence and show TOP and the stack contents after each operation.

C. Queue Tracing Drills (with answers)

Drill 1 — Linear queue: Capacity 5; enqueue 10, 20, 30, 40; dequeue twice; enqueue 50 and 60. Under the standard simple-array implementation without shifting, insertion can fail when REAR reaches the last index even though earlier slots are empty. This is called false overflow.

Drill 2 — Circular queue: Capacity 5; enqueue 10, 20, 30, 40; dequeue twice; enqueue 50, 60, 70. Track FRONT and REAR with modulo-5 wraparound; use the chosen implementation's full condition consistently.

Drill 3 — Ascending priority queue: Insert 25, 10, 30, 15; if the smallest value has highest priority, deletion order is `10, 15, 25, 30`.

Drill 4 — Deque: insertRear(10), insertFront(20), insertRear(30), deleteFront(), deleteRear(). Final deque contains only `10`.

Part V — Mock 50-Mark Paper

Part A — 10 marks

QA1. Classify data structures and give examples. (5 marks)

QA2. Define a queue and explain its basic operations. (5 marks)

Part B — 20 marks (attempt one alternative)

Q2. (a) Write an infix-to-postfix conversion algorithm. (5)
(b) Convert `A+B*(C-D)` to postfix. (5)
(c) Explain and trace a recursive function with a suitable example. (10)

OR

Q3. (a) Differentiate linear and circular queues. (5)
(b) Write circular queue insertion and deletion algorithms. (5)
(c) Trace circular queue operations for capacity 5. (10)

Part C — 20 marks (attempt one alternative)

Q4. (a) Explain stack operations and array representation. (5)
(b) Evaluate a postfix expression using a stack. (5)
(c) Trace recursion, showing calls and return values. (10)

OR

Q5. (a) Explain queue types and applications. (5)
(b) Explain ascending and descending priority queues. (5)
(c) Write deque insertion/deletion algorithms. (10)

Final Revision Priority

1. Infix-to-postfix conversion, precedence/associativity and stack trace.

2. Circular queue: array trace, modulo arithmetic, full/empty conditions and algorithms.

3. Postfix evaluation and recursion/function tracing.

4. Priority queues and deques: types, algorithms and applications.

5. Introductory data-structure theory: classification, operations and comparisons.

Scope reminder: This bank stops at queues in Unit 2. Linked lists and later topics are intentionally excluded. The online PYQ patterns mentioned in the conversation were from earlier BMSCE papers/curriculum versions, so use them to guide practice rather than as a prediction of the exact upcoming paper.